

№₀ Weekly Problem

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Problem

Let a_n be the number of terms in the sequence $2^1, 2^2, \dots, 2^n$ which begin with digit 1. Prove that

$$\log 2 - \frac{1}{n} < \frac{a_n}{n} < \log 2,$$

and, hence,

$$\lim_{n \rightarrow \infty} \frac{a_n}{n} = \log 2 \approx 0.30103.$$

Solution

Proof. We first prove the following lemma, greatly simplifying the remainder of the proof.

Lemma 1. Define $(s_k) = (2^j)_{j=1}^k$, the sequence of the first k integer powers of 2. Also, let S_c be the set of such sequences all having exactly $c \in \mathbb{N}$ terms whose leading digit is 1. If $(s_m) \in S_c$, then $c \log_2 10 < m < (c+1) \log_2 10$.

Proof. Suppose that $(s_m) \in S_c$ for some $c \in \mathbb{N}$. We will show that $c \log_2 10 < m < (c+1) \log_2 10$.

Observe that for $2^q \in \mathbb{R}$ to have a leading 1, there must exist some $t \in \mathbb{N}$ such that

$$10^t \leq 2^q < 2 \cdot 10^t \implies t \log_2 10 \leq q < t \log_2 10 + 1.$$

For fixed t , the interval $[t \log_2 10, t \log_2 10 + 1)$ has length 1, so q can only take on, at most, one integer value.

Let $j_1 < j_2 < \dots < j_c \leq m$ be the exponents of the terms of (s_m) that have a leading 1. Therefore, each $j_i \in \mathbb{N}$ is contained in mutually exclusive unit intervals $[t_i \log_2 10, t_i \log_2 10 + 1)$ and $t_1 < t_2 < \dots < t_c$. Also, $t_1 \geq 1$ since $2^{j_1} = 2^4 = 16$ is the first integer power of 2 with 1 as its most significant digit. Because $(t_i)_{i=1}^c$ is strictly increasing, we have $t_i \geq i$ for all $1 \leq i \leq c$. Thus, for all $1 \leq i \leq c$, yielding

$$j_i \geq t_i \log_2 10 \geq i \log_2 10.$$

In particular, since $\log_2 10$ is irrational, we have shown that

$$m \geq j_c > c \log_2 10 \implies m > c \log_2 10.$$

To establish the upperbound (i.e., $m < (c+1) \log_2 10$), we proceed in a similar fashion. Consider the next exponent j_{c+1} wherein $2^{j_{c+1}}$ starts with a 1. Clearly, $m < j_{c+1}$, and much of same reasoning above holds:

1. $\exists t_{c+1} \in \mathbb{N}. t_{c+1} \log_2 10 \leq j_{c+1} \leq t_{c+1} \log_2 10 + 1$
2. $t_1 < t_2 < \dots < t_c < t_{c+1}$
3. $t_{c+1} \geq c+1$.

Now suppose to the contrary that $t_{c+1} \geq c+2$. This would imply $j_{c+1} \notin [t_{c+1} \log_2 10, t_{c+1} \log_2 10 + 1)$, or equivalently, that this interval is devoid of an integer power of 2. But, this gives us our desired contradiction: there exists $q \in \mathbb{N}$ such that $2^q \in [t_{c+1} \log_2 10, t_{c+1} \log_2 10 + 1)$ contradicting the fact that j_{c+1} is the next exponent where $2^{j_{c+1}}$ starts with a 1. Hence, we have $t_{c+1} = c+1$, producing

$$m < j_{c+1} < (c+1) \log_2 10 + 1.$$

Finally, since $\log_2 10$ is irrational and $m \leq j_{c+1} - 1$ (m, j_{c+1} are integers and $m < j_{c+1}$), we have demonstrated that

$$m \leq j_{c+1} - 1 < (c+1) \log_2 10 \implies m < (c+1) \log_2 10.$$

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Turning our attention to the original problem, it is sufficient to show that

$$a_n \log_{10} 2 < n < (a_n + 1) \log_{10} 2$$

for all $n \in \mathbb{N}$ because

$$\begin{aligned} \log 2 - \frac{1}{n} < \frac{a_n}{n} < \log 2 &\iff n \log 2 - 1 < a_n < n \log 2 \\ &\iff \frac{n}{\log_2 10} - 1 < a_n < \frac{n}{\log_2 10} \\ &\iff a_n \log_2 10 < n < (a_n + 1) \log_2 10. \end{aligned}$$

Let $n \in \mathbb{N}$ be arbitrary. Then, (s_n) has a_n terms whose leading digit is 1 and $(s_n) \in S_{a_n}$. By Lemma 1,

$$a_n \log_2 10 < n < (a_n + 1) \log_2 10,$$

as desired. Since n is arbitrary, we can write

$$\forall n \in \mathbb{N}. \log 2 - \frac{1}{n} < \frac{a_n}{n} < \log 2.$$

By the Squeeze Theorem, we immediately obtain

$$\lim_{n \rightarrow \infty} \frac{a_n}{n} = \log 2 \approx 0.30103.$$

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